

Compulsory exercise, Algorithms number 1

Due date: September 4th 2026

(At most three students per group. Does **not** count in the final grade)

Problem 1 (Brute-force)

Draw a figure illustrating the comparisons done by the brute-force pattern matching algorithm for the case when the text is "aaabaadaabaaa" and the pattern is "aabaaa". How many comparisons of characters are done by the algorithm?

Problem 2 (Boyer-Moore)

- a) Repeat the previous problem for the BM matching algorithm, not counting the comparisons made to compute the last function.
- b) The theory says that experiments shows that on English text, the average number of comparisons done per text character is approximately 0.24 for a five character pattern string. Implement the algorithm in **Python** and run it on a Norwegian (or another language) text and compare the results. The reason for using Python is that the Machine learning part of the course use Python.

Problem 3 (Knuth-Morris-Pratt)

Repeat a) in the previous problem for the KMP matching algorithm, not counting the comparisons made to compute the failure function.

Problem 4

How many nonempty prefixes of the string $P = \text{"aaabbaaa"}$ are also suffixes of P ?

Problem 5

Draw a standard trie for the following set of strings

{abab, baba, cccc, baaaa, caa, bbaacc, cbcc, cbca}

Problem 6

Draw a compressed trie for the set of strings of the previous problem

Problem 7

Draw the frequency table and Huffman tree for the following string

"dogs do not spot hot pots or cats"

Problem 8

Show how to use dynamic programming to compute the number of characters in the longest common subsequence between the two strings "babbabab" and "bbabbaaab".

Problem 9

- a) Implement a recursive version in Python of the longest common subsequence problem.
- b) Implement the dynamic version of the longest common subsequence problem in Python.
- c) Experiment with strings of different length. At some point the recursive version will start using "a lot of time" while the dynamic version is still fast. Approximately, at what point does this happen?